

A Novel Implementation of QPSK Modulator on FPGA

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Abstract—QPSK is one of the digital modulation techniques in communication system that has wide applications in the field of wireless, satellite communication, Bluetooth, OFDM etc. The Conventional QPSK modulators designs are generally based on IQ approach. In these modulators multipliers are used to separate the modulating signal into the in-phase and quadrature phase which increases the hardware complexity. The objective of this work is to implement the digital QPSK modulator with conventional and another approach of digitization. In this approach a digitized carrier wave is generated and QPSK phase shifting is provided by 4:1 multiplexer according to the data bits. The implementations of both the designs are done on Spartan 3E Xilinx FPGA through the verilog hardware description language. The QPSK modulator based on the Digitized approach is simpler in terms of hardware circuitry.

Keywords— Quadrature Phase Shift Keying (QPSK); Deterministic Random Bit Generator (DRBG); Verilog; Look up table (LUT); FPGA; Serial in Parallel out (SIPO).

I. INTRODUCTION

Modulation is phenomena in which characteristic like amplitude, phase and frequency of a high frequency carrier wave is changed accordingly low frequency modulating wave. Modulation is required to minimize the noise in signal and to separate the signal from different transmitters. Generally, Modulation can be done by two ways the analog and digital modulation. Digital modulation provides more information capacity, greater noise immunity, robustness to channel impairments, higher data security, better quality communication, compatibility with digital data service, faster system availability and easier multiplexing of various forms of information like voice, data and video etc.

This modulation technique is used to send the computer information over transmission channels that require analog signals like a fiber optic network, computer modems, cellular phone network, and satellite systems. In these types of system, an electromagnetic wave is used to modulate the information over long distances and connect digital information user at remote locations. One or more parameter of the carrier wave is modulated by the digital data, this process is called “shift – keying” to differentiate it from the purely analog systems like AM and FM. The most common modulation technique used in digital communication system is quadrature phase shift keying

(QPSK) which is the type of phase shift keying modulation. It is most simple but robust form of Digital Modulation techniques. It is used in Satellite Communication, Wireless Communication etc. In QPSK data bits are grouped into pairs and each pair is represented by a particular waveform called a symbol to be sent across the channel after modulating the carrier. Better accuracy of the transmitted digital signal is measured by “Bit Error Rate (BER)”. There is no bit error rate (BER) degradation in QPSK while the bandwidth efficiency is increased [1].

II. CONVENTIONAL QPSK MODULATOR

In Quadrature Phase Shift Keying (QPSK), the data bits to be modulated are paired into symbols each containing two bits [2]. So, each symbol can take the one of four possible values 00, 01, 10, 11. Fig. 1 shows the block diagram for the Conventional QPSK modulator. The circuit is based on IQ modulation [3]. As the symbol contains two bits, so one bit modulates in phase-carrier and second bit will modulate the quadrature phase-carrier. The bits are named as the I and Q respectively, and are transmitted to the modulator via a Serial-In-Parallel-Out (SIPO) shift register. SIPO accepts the input from the data and converts it to the parallel data.

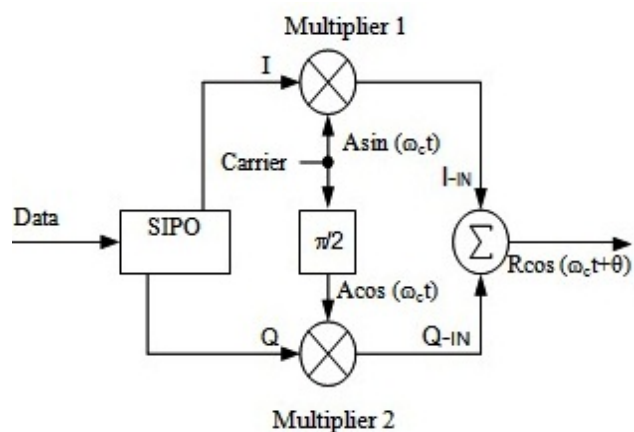


Figure 1. Conventional QPSK modulator

Fig. 1 contains the two multipliers; each multiplier used in the circuit takes the input +1 and -1 respectively according to

the 1 and 0 of I-Q bits. In multiplier 1 the I-bit and carrier $\text{Asin}(\omega_c t)$ is multiplied and the multiplier 2 the Q-bit and carrier with phase 90° $\text{Acos}(\omega_c t)$ is multiplied. For Shifting the carrier by 90° phase shifter is used. The value of I, Q, I-IN and Q-IN with all combination of I and Q are shown in the table I.

TABLE I. SIGNAL VALUES OF IQ QPSK MODULATOR

I	Q	I-IN	Q-IN
0	0	$-\text{Asin}(\omega_c t)$	$-\text{Acos}(\omega_c t)$
0	1	$-\text{Asin}(\omega_c t)$	$\text{Acos}(\omega_c t)$
1	0	$\text{Asin}(\omega_c t)$	$-\text{Acos}(\omega_c t)$
1	1	$\text{Asin}(\omega_c t)$	$\text{Acos}(\omega_c t)$

So, the QPSK out will be the sum of two incoming inputs that is I-IN and the Q-IN.

$$I \cdot \text{Asin}(\omega_c t) + Q \cdot \text{Acos}(\omega_c t) = R \cos(\omega_c t + \theta) \tag{1}$$

$$\text{Where } R = \sqrt{I^2 + Q^2}$$

$$\theta = \tan^{-1}\left(\frac{Q}{I}\right)$$

In equation “(1)”, the R will be $\pm A/\sqrt{2}$ where A is the amplitude of the carrier and θ will be -135° , $+135^\circ$, -45° and $+45^\circ$ according to the symbol 00, 01, 10 and 11. The resulting QPSK output will contain four phase’s, they will be 90° apart from each other.

III. PROPOSED QPSK MODULATOR

The new approach is proposed for QPSK modulator. In this method the QPSK is generated on the digital platform by means of digitized carrier wave generator [4]. The Proposed circuit does not use the multipliers and mixers but the conventional IQ modulator uses two multipliers, so the proposed circuit minimizes the circuit area as well as cost.

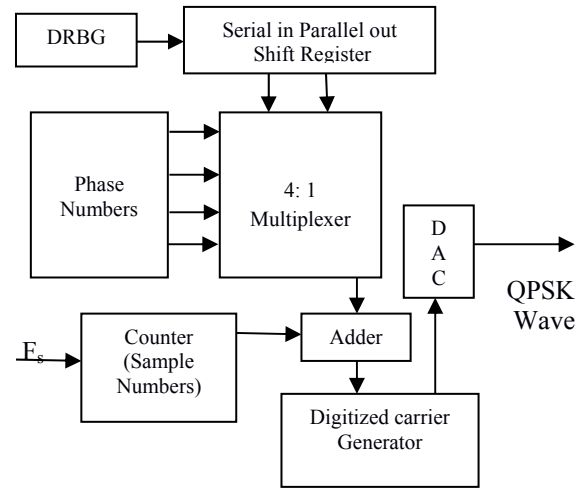


Figure 2. Novel QPSK modulator

The overall design process is divided into four steps is shown in Fig. 2, first as for simulation process Deterministic Random Bit Generator (DRBG) is used to produce different combination of inputs [5].

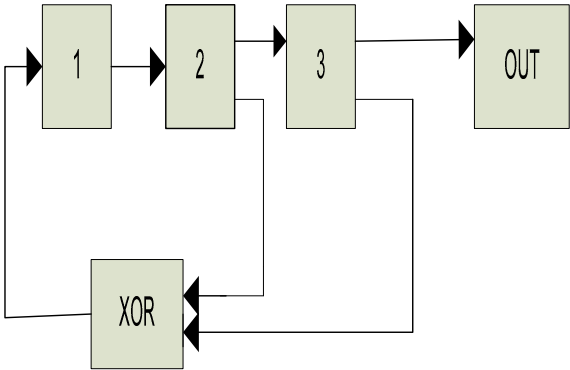


Figure 3. Deterministic Random Bit Generator

The DRBG is an algorithm for generating a sequence of random bits; the sequence is not completely random in DRBG it can be determined by small set of initial values. The data bits are generated from the DRBG shown in Fig. 3, the bit sequence is generated with the help of the XOR gate and the generated output is sent as the data bits.

In the second step the serial bit sequence will be the input for the serial in parallel out shift register; this register converts the DRBS input to parallel bits [6]. In QPSK the carrier undergoes four changes in phase (four symbols) and can denote two binary bits of data per symbol, and each symbol differ by one bit only, so it follows the gray encoded bit pattern. The phase changes according to input data sequence in QPSK [4].

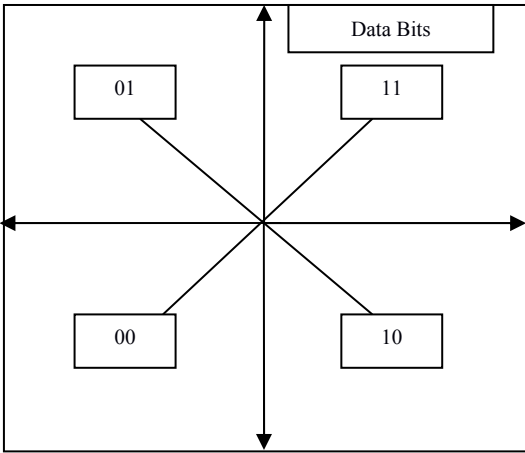


Figure 4. QPSK Constellation diagram

The phase in QPSK changes according to the constellation diagram shown in Fig. 4, here it is shown that change in one bit in the input changes the phase by 90° and in proposed method we decide the phase number according to the phase shift. A 4:1 Multiplexer is used to obtain the particular phase number according to the symbols shown in the table II.

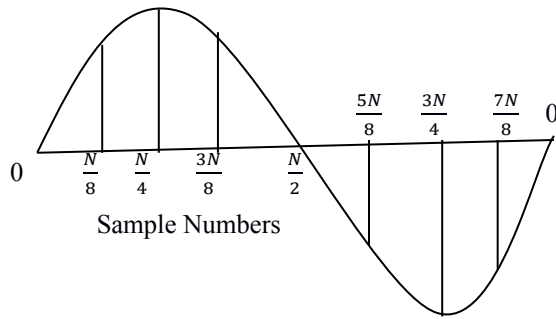


Figure 5. Phase numbers by constellation diagram

From the above Fig. 5 we can decide the phase numbers according to bits sequence, Let if N samples are taken thus the Phase numbers corresponding to the phase shift -135° , $+135^\circ$, -45° and $+45^\circ$ will be $5N/8$, $3N/8$, $7N/8$ and $N/8$ respectively.

The generated phase number provides the four phase shifts required for QPSK modulation. The 4:1 Multiplexer input for the phase shift with input sequence and phase number is shown below in table II.

TABLE II. PHASE SHIFT FOR QPSK

Input	Phase Shift	Phase Number(Binary)
00	-135°	$5N/8$
10	-45°	$7N/8$
11	45°	$N/8$
01	135°	$3N/8$

In the step third the counter is designed at sampling frequency F_s to generate the sample number. The phase number generated is combined with the sample generator by means of adder. After by means of LUT the digitized Carrier wave is generated according to the output of adder. In the final step the output of digitized carrier wave is converted in analog form by help of digital to analog converter (DAC), so the QPSK Modulation is obtained.

IV. DISCUSSION ON RESULTS

Both the conventional and proposed QPSK Modulator is designed implemented on FPGA through verilog hardware description language [7]. The Spartan 3E Starter kit device XC3C500E-4FG320 was used for implementation and simulated on modelsim –SE verilog [8]. In the below Fig. 6 simulated timing waveforms is obtained from modelsim simulator which signifies the result of QPSK_out waveform.

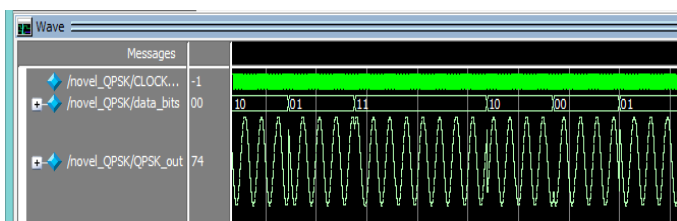


Figure 6. Simulation results of QPSK Modulation

After the FPGA implementation the final output has seen on the digital oscilloscope and the DSO picture has shown in Fig. 7.



Figure 7. QPSK Modulated waveform on DSO

So, both the IQ QPSK Modulator and digitized QPSK Modulator are implemented on FPFA but now we compare both the methods in terms of hardware utilized in the design process and size. So here is the flow summary for the hardware utilization in Fig. 8 for IQ QPSK Modulator.

Device Utilization Summary				
Logic Utilization	Used	Available	Utilization	Note(s)
Number of Slice Flip Flops	62	9,312	1%	
Number of 4 input LUTs	119	9,312	1%	
Number of occupied Slices	79	4,656	1%	
Number of Slices containing only related logic	79	79	100%	
Number of Slices containing unrelated logic	0	79	0%	
Total Number of 4 input LUTs	141	9,312	1%	
Number used as logic	119			
Number used as a route-thru	22			
Number of bonded IOBs	13	232	5%	
Number of BUFGMUXs	2	24	8%	
Average Fanout of Non-Clock Nets	2.59			

Figure 8. IQ QPSK Modulator device utilization

Fig. 9 shows the Device utilization summary for Proposed QPSK Modulator implementation.

Device Utilization Summary				
Logic Utilization	Used	Available	Utilization	Note(s)
Number of Slice Flip Flops	25	9,312	1%	
Number of 4 input LUTs	30	9,312	1%	
Number of occupied Slices	26	4,656	1%	
Number of Slices containing only related logic	26	26	100%	
Number of Slices containing unrelated logic	0	26	0%	
Total Number of 4 input LUTs	46	9,312	1%	
Number used as logic	30			
Number used as a route-thru	16			
Number of bonded IOBs	13	232	5%	
Number of BUFGMUXs	1	24	4%	
Average Fanout of Non-Clock Nets	3.04			

Figure 9. Novel QPSK Modulator device utilization

By comparing both the design summary we can easily analyze that how much hardware saved by using the digitized QPSK Modulator. So here is the table III for hardware comparison.

TABLE III. HARDWARE COMPARISON SUMMARY

Logic Utilization	IQ	Digitized	Difference (%)
Slice FF	62	25	>50
4 i/p LUT	119	30	>70
Occupied slices	79	26	>70
Slices with related logic	79	26	>70
Slices with unrelated logic	0	0	-
Total no. of 4 i/p LUT	141	46	>70
Bonded IOBs	13	13	same
No. of BUFGMUX	2	1	50

So from above table we can see that the hardware used for the proposed technique is really less in comparison to the conventional IQ Modulator.

V. CONCLUSION

This paper presents implementation of the IQ based QPSK Modulator and the proposed digitized QPSK Modulator using Spartan 3E starter Xilinx Board. The board contains rich set of features and professionals to a wide range of topics for FPGA design implementation as well as for the development of sophisticated digital systems by using verilog hardware description language.

In this paper it is tried to achieve a design for QPSK modulation which will give the same result by minimizing its size or area as well. The reduced size of the design will reduce the power consumption. The proposed block diagram replaces several blocks like multiplier and mixers which will help to increase speed of operation of the design. It means proposed block diagram try to improve all the three main factors power, area, hardware which are taken into consideration for any system design.

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